

Domain Randomization for Robust Sampling-Based Model Predictive Control on Bipedal Wheeled Robots

Motivation:

Reinforcement learning (RL) with domain randomization (DR) enables robust sim-to-real transfer for legged robots [1, 2], but produces fixed policies that cannot adapt online. GPU-accelerated sampling-based MPC [3, 4, 5, 6] offers online adaptability and interpretable objectives, yet suffers from **brittle sim-to-real transfer** requiring extensive manual tuning [4]. Traditional robust MPC (tube MPC, H-infinity) struggles with contact-rich dynamics [7, 8], and DR has not been systematically studied for sampling-based MPC.

We target the LIMX Tron1, a bipedal wheeled robot (8 DoF: 3 per leg + 1 per wheel) that is complex enough to stress-test robustness yet compact enough for thorough ablation. Crucially, sampling-based MPC *already* evaluates thousands of rollouts in parallel; randomizing dynamics parameters across these rollouts adds minimal cost, transforming each MPC step into an implicit distributionally robust optimization. This can be further accelerated with learned surrogate dynamics [9, 10]. If DR can close the robustness gap, it would combine RL’s sim-to-real reliability with MPC’s online adaptability.



Goal:

We aim to integrate domain randomization directly into the sampling-based MPC loop for robust sim-to-real transfer on the LIMX Tron1, without requiring online parameter estimation or per-robot tuning. Concretely:

- Build a DR-MPC framework that randomizes dynamics parameters (friction, mass, actuator gains, contact properties) across GPU-parallelized rollouts using physics simulators or learned surrogates [11, 12]
- Ablate key design choices: parameter selection, randomization range, and cost aggregation (mean, worst-case, CVaR [13, 14])
- Show that DR-MPC achieves sim-to-real robustness competitive with RL while retaining MPC’s online adaptability

Approach:

We propose two complementary instantiations:

(1) DR-MPC with Physics Simulation. Building on DIAL-MPC [4] or Hydrax [5] with MuJoCo MJX [15], we randomize dynamics parameters across parallel rollouts and select actions via MPPI/CEM [16, 17] that minimize cost across the parameter distribution. Cost aggregation ranges from mean (risk-neutral) to CVaR for tunable risk sensitivity [18].

(2) DR-MPC with Learned Surrogate Dynamics. We train a neural dynamics model conditioned on physical parameters (mass, friction, stiffness, latency) on domain-randomized data [9, 10, 19], replacing full physics simulation with cheap forward passes. Lipschitz regularization ensures smooth rollouts, and GPU-parallel execution leverages CusADi [20] or GATO [21]. The Tron1’s compact 8-DoF state space makes surrogate learning feasible; a hierarchical architecture [22] can further decompose planning across timescales.

Ablation axes: parameter selection (friction, mass, actuator gains, damping), randomization range (5%–30%), cost aggregation (mean, minimax, CVaR), and sample budget split between action and parameter diversity [23].

Evaluation. Tasks include velocity tracking, push recovery, slope traversal, and turning on the LIMX Tron1. Baselines: nominal MPC [4], DR-trained RL [1, 24, 25], and analytical MPC [7]. Metrics: robustness under perturbations, tracking accuracy, energy, planning frequency (at least 50 Hz), and online adaptability. Hardware validation is planned pending access.

General Details:

The student should bring along the following attributes:

1. Proficiency in Python and machine learning (familiar with PyTorch/JAX).
2. Good knowledge of model predictive control and/or reinforcement learning.
3. Experience with physics simulation (MuJoCo, Isaac Gym) is preferred.
4. Familiarity with GPU-accelerated computing is a plus.

Interested?

Reach out to Jin Cheng (jin.cheng@inf.ethz.ch) and Valentin N. Hartmann (valentin.hartmann@inf.ethz.ch) with your CV and transcripts.

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